Predicting Veterinarians Licensure Exam Success

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Problem & Data Overview

Objective

Predict licensure exam success (Pass/Fail binary outcome) using student performance data

Dataset

Student-level aggregated mock exam data with comprehensive performance metrics

Features

Scores, attempts, trends, and subject-level performance indicators

Target Variable

Licensure Result (Pass/Fail) with data leakage removed

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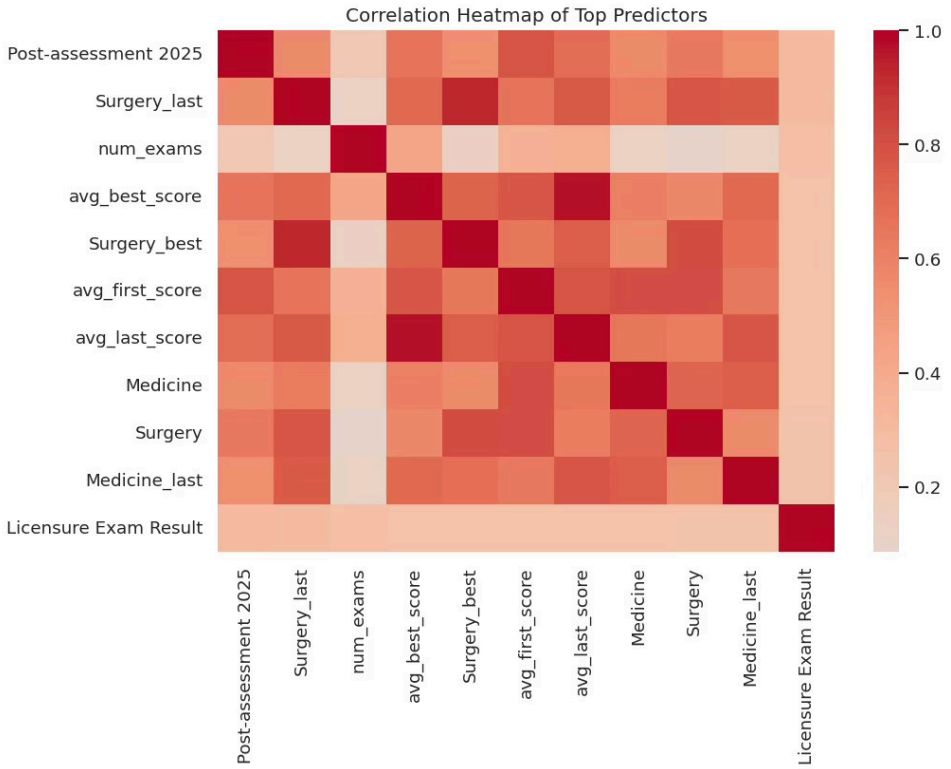
EDA Key Findings

Strong Correlations

Mock exam scores show **strong correlation** with licensure outcome, particularly average and best scores.



Score Performance

Higher average and best scores correlate with **increased probability** of passing licensure exams

Consistency Matters

Low score variance and **consistent performance** improve success likelihood significantly

Repeaters' Patterns

Students with multiple attempts show **distinct performance patterns** that aid prediction

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Feature Engineering Approach

Score Aggregation

First, last, and best scores extracted from all attempts

Trend Analysis

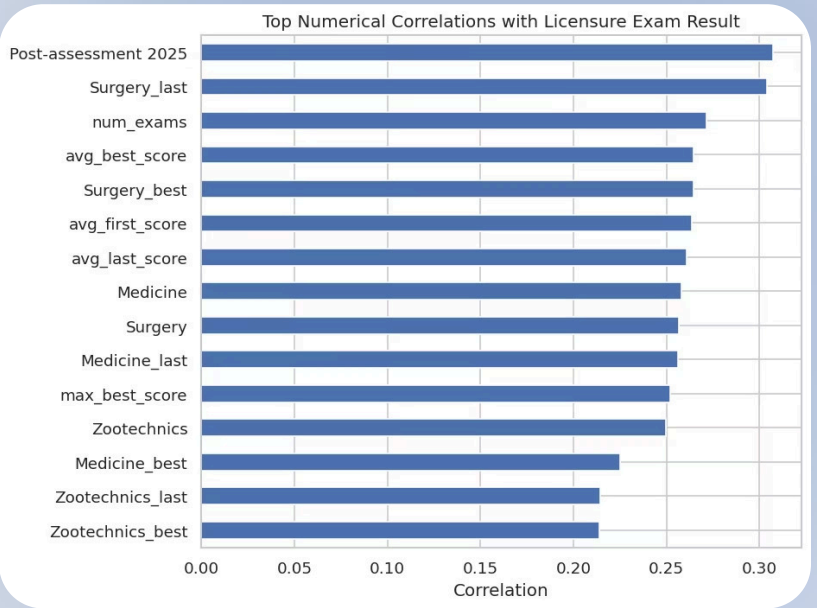
Improvement metrics calculated as best minus first score

Stability Metrics

Standard deviation and consistency index measured

Engagement

Total attempts and exam coverage tracked



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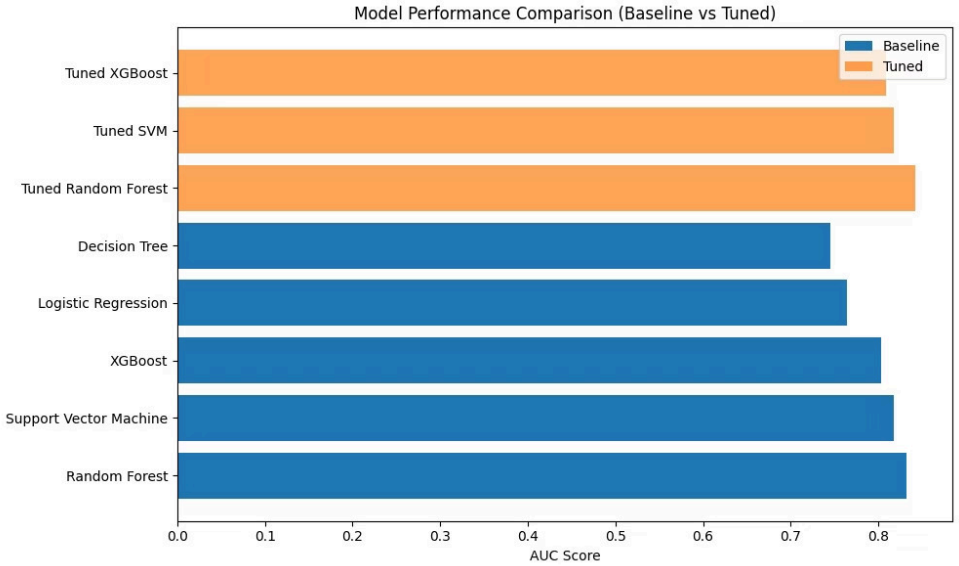
Model Selection & Methodology



Models Evaluated

- Logistic Regression
- Support Vector Machines
- Decision Tree
- Random Forest
- XGBoost

01

Train-Test Split

Standard train-test split (80:20) applied for generalization

02

Hyperparameter Tuning

Randomized search and cross-validation applied to optimize parameters

03

Evaluation Metrics

Accuracy, ROC-AUC, Precision, Recall calculated

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Results & Model Comparison

Best Model

Random Forest achieved highest performance

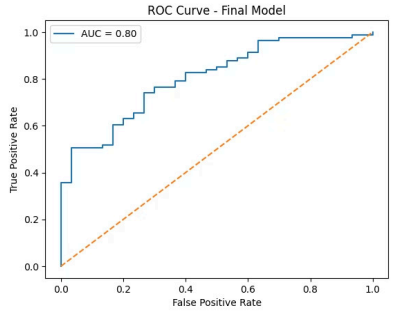
ROC-AUC Score

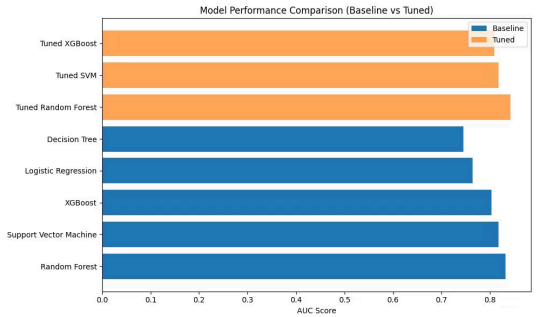
≈0.80 indicates strong predictive power

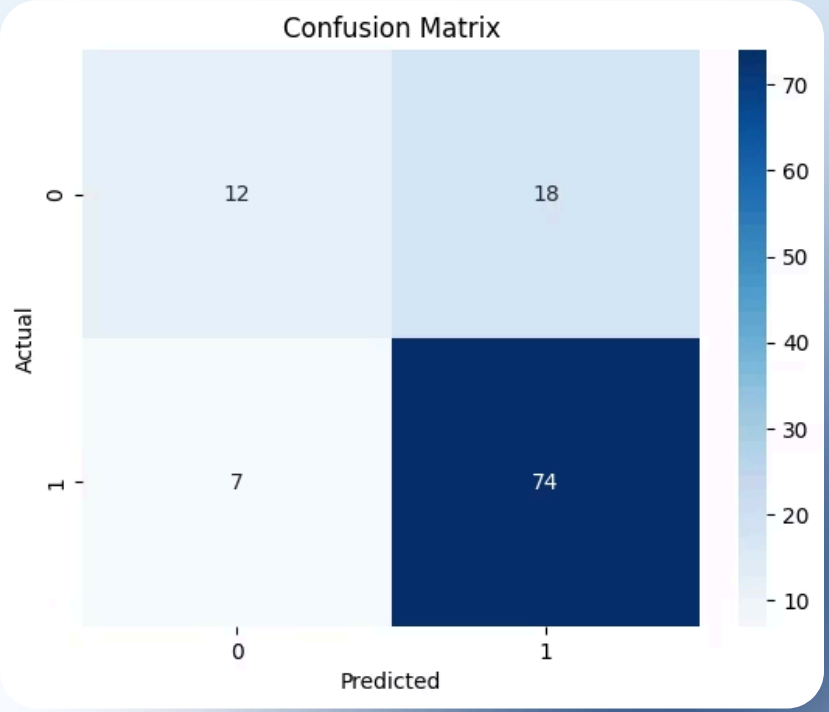
Balanced Performance

Good balance between precision and recall

Performance Visualization







Model suitable for early warning system to identify at-risk students for targeted interventions.

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Feature Importance & Explainability

Interpretability Methods

SHAP framework applied for **model interpretability** and actionable insights.

1

Average First Score

Strong baseline predictor of ultimate success

2

Zootechnics Score

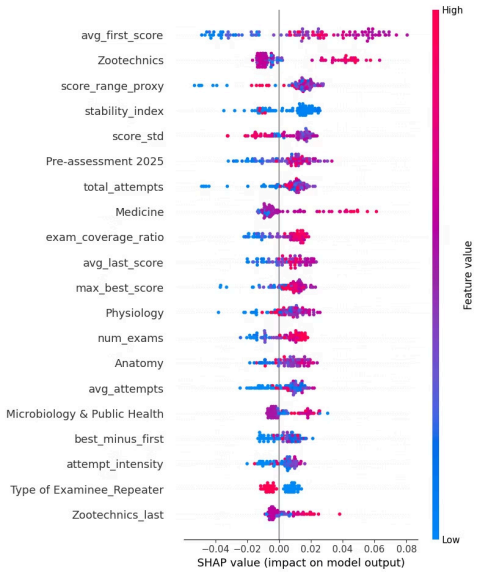
Subject-specific performance as key indicator

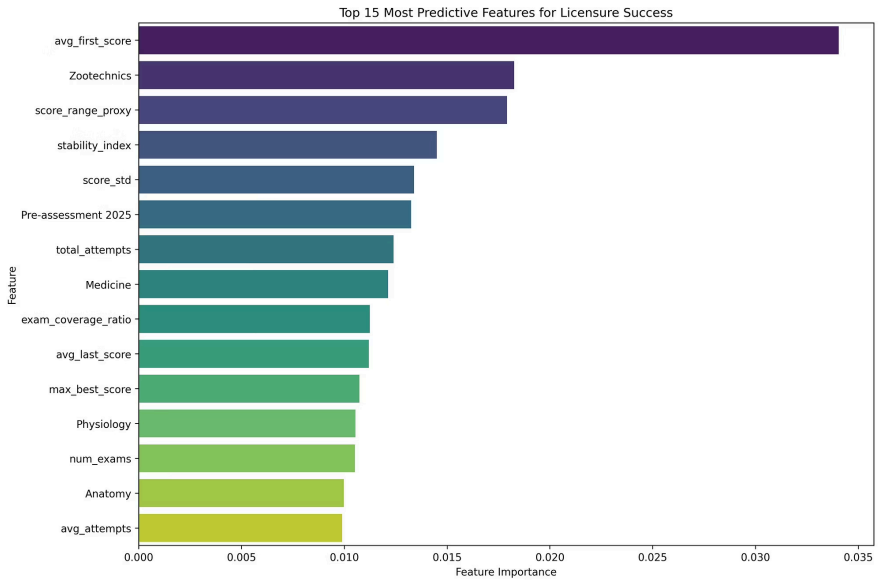
3

Score Range Proxy

Consistency and performance trends as critical drivers

SHAP Summary Plot





Model decisions are explainable and actionable for educational interventions.

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Limitations & Future Work

Limited Dataset Size

Current dataset (n=552) may affect **model generalization** to broader populations

Potential Bias

Need to monitor for **systematic bias** across different student groups

Model Retraining

Continuous retraining required as **new data becomes available**

Future Enhancements

- **Real-time data integration** for immediate predictions
- **Adaptive recommendation systems** for personalized interventions

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